

The Reduction of Universality: A Regime Boundary Theory for Finite-System Phase Transitions

—Cross-Model Evidence for Bimodal Splitting, the Precursor Regime Boundary, and the Tool-Dependence Theorem

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Abstract

Central proposition: Traditional universality is a mathematical convergence artifact of the thermodynamic limit ($L \rightarrow \infty$). In real, finite systems, regime boundaries are more fundamental physical quantities than universality classes. Using the Factor Hierarchy Law and the Testability Norms (six functional dimensions $\times$ six-tool cross-validation), we performed a systematic cross-model analysis on five phase-transition models (2D Ising exact solution, 3D Ising, 2D Potts $q=3$, 2D Potts $q=4$, and 2D XY). Five core discoveries are reported:

(1) Bimodal splitting. In a finite lattice, a second-order phase transition splits into two successive regime switches—locking of the ground state (M_{sp} dimension) and a structural change of fluctuations (U_4 dimension)—separated by an independent transition regime. On 3D Ising $L=64$, M_{sp} breaks at $T \approx 4.40$ – 4.50 (Chow $F=939.9$ – 1524.1 , depending on scan parameters) and U_4 breaks at $T \approx 4.50$ – 4.55 ($F=299.1$ – 2335). The same structure is reproduced on 2D Potts $q=4$ for $L=16, 32$, and 64 . Blind Bai-Perron double-breakpoint detection, independent of the Chow-scan parameter, confirms the bimodal-splitting structure directly from the M - h coefficient series.

(2) Precursor regime boundary. The breakpoint of the M_{sp} dimension across $L=16$ – 128 exhibits a mean of 4.50 and a standard deviation of only 0.071 —almost immobile despite a 512 -fold increase in volume. Potts $q=4$ ($L=16$ – 128) independently reproduces this stability. In contrast, the KT transition (XY model) displays a strictly monotonic convergence ($p=0.0000$). The available data ($L \leq 128$) are not yet sufficient to statistically distinguish "slow convergence" from "structural non-convergence", but the sharp contrast in size-sensitivity between the two transition types constitutes a robust empirical regularity.

(3) Tool-dependence theorem. On the exact 2D Ising solution ($T_c=2.269185$), the Chow test ($T=2.2735$) and the susceptibility-peak method ($T=2.2675$) give "critical temperatures" lying on opposite sides of the true value, differing by 0.006 . On 3D Ising $L=64$, five methods applied across multiple observables yield a T_c range spanning 0.325 temperature units. No "critical temperature" exists independently of the detection tool.

(4) χ -blindness. On the exact 2D Ising solution, χ yields a Chow F of only 16.8 , whereas $1/\chi$ reaches $\sim 10^{21}$; on 3D Ising $L=64$, the activation factor is 40 . The root cause is a mathematical mismatch between the even-function structure of χ

and the odd-function requirement of the Chow test—fifty years of locating T_c via the χ peak have systematically missed the "regime-switch" dimension of phase transitions.

(5) Random-bond Griffiths-phase verification. Traditional universality theory predicts that pure and disordered models are equivalent in the long-wavelength limit. The Factor Hierarchy Law, however, predicts that disorder will generate additional regime boundaries. On a 10% disordered 3D Ising model, Bai-Perron detects a new breakpoint at $T \approx 4.25$ that is absent in the pure model (significant on both $L=16$ and $L=32$, with a BIC improvement up to +42.77). CUSUM and MOSUM are uniformly significant (permutation test $p=0.0000$). This result constitutes decisive evidence for a paradigm shift—where traditional theory makes an incorrect prediction, the Factor Hierarchy Law makes the correct one.

Based on the above, we establish an MRSD tripartite classification framework: first-order transitions display mixed-type switching (intercept jump and slope change of comparable magnitude) in finite systems, second-order transitions display rule-resetting + bimodal splitting, and KT transitions display single-stage continuous crossover. A modified finite-size scaling ansatz is proposed: $B(L) = T_c(\infty) \pm \Delta T_0 \cdot \Phi(L)$, where $\Phi(L) \approx \text{constant}$ for certain regime boundaries over an intermediate range of scales. Overall conclusion: Universality is a convergence product of the thermodynamic limit. In real finite systems, regime boundaries are more fundamental than universality classes. This paradigm shift, from "universality" to "regime boundaries", constitutes a "reduction of universality" within the traditional theory of phase transitions.

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1. Introduction

1.1 A century-old paradigm and its fatal boundary

Since Ehrenfest (1933) proposed the classical classification of phase transitions, physics has relied on a fundamental premise: one must be able to write down the free-energy function of the system. First-order transitions are defined by a discontinuous jump in the first derivative of the free energy; second-order transitions by a divergence or jump in the second derivative. Combined with the renormalization group (Wilson, 1971), this framework gave birth to the concept of "universality"—critical exponents depend only on spatial dimensionality and the symmetry of the order parameter, not on microscopic details.

Yet this paradigm has a fatal boundary. The partition function of the 3D Ising model—the standard model for real ferromagnets—was rigorously proved to be NP-complete (Istrail, 2000). The pairing

mechanism of high-temperature superconductors, glass transitions in non-equilibrium systems—these fields produce vast amounts of data for which the Ehrenfest classification offers no guidance.

A deeper problem exists: even on exactly solvable systems, universality holds only in the thermodynamic limit ($L \rightarrow \infty$). In any real, finite system, the partition function is a finite sum of analytic exponentials—and a finite sum of analytic functions is itself analytic. Hence, a finite system cannot possess non-analytic singularities or genuine divergences. Universality is an ideal in the mathematician's hands, not a reality in the experimenter's.

1.2 The Factor Hierarchy Law and the Testability Norms

Tang (2026g) proposed the Factor Hierarchy Law: in any complex system driven by multiple quantifiable factors, factors naturally fall into two tiers—regime factors and executive factors. A regime factor determines which set of rules is currently active; its function is not reflected in direct explanatory power but in how it modulates the exposure coefficients of executive factors through interaction effects. When a regime factor crosses a critical value, the weights of all factors in the system are systematically reset. This law has been independently verified across five major financial markets (Tang, 2026a–2026f) and five independent dimensions in astronomy (Tang, 2026h, 2026i, 2026j).

The Testability Norms (Tang, 2026g) are the operational realization of the law, comprising six functional dimensions—search completeness, modulation relation, structural break, causal direction, problem completeness, and process closed-loop—together with a set of replaceable implementation tools. Tang (2026k) proved the exact information-theoretic duality between the Factor Hierarchy Law and the Ehrenfest classification on the exactly solved 2D Ising model: the Chow test attains a unique global maximum at the exact T_c , and the deviation is proved analytically to be strictly zero. Tang (2026l) discovered hidden structures of finite-lattice phase transitions on the 3D Ising model (an NP-complete system)—bimodal splitting, the transition regime, and the precursor regime boundary—and distilled the Multi-Regime State Decoupling (MRSD) algorithm.

1.3 Core tasks and argument logic of the present paper

This paper has a dual objective. First, to perform a complete cross-model verification of the Testability Norms on five phase-transition models and to establish an MRSD tripartite classification of finite-system phase transitions. Second, to argue, on the basis of cross-model evidence, the central

proposition: **Traditional universality is a mathematical convergence product of the thermodynamic limit; in real finite systems, regime boundaries are more fundamental than universality classes.**

The argument follows a clear logical chain, which constitutes a "reduction" operation on the concept of universality:

1.Tool dependence (§3): We prove that "critical temperature" is not an objective physical quantity but a function of the detection tool. If T_c itself depends on the choice of tool, the whole notion of "universality" built upon it must be re-examined.

2.Discovery of regime boundaries (§4–5): We demonstrate that finite systems possess intrinsic structural boundaries (bimodal splitting, the precursor regime boundary) that do not vanish with increasing system size and do not converge to the thermodynamic limit.

3.Failure of the traditional theory (§6): In a random-bond model, traditional universality treats the pure and disordered models as equivalent, whereas our framework detects regime boundaries absent from the pure model—the traditional theory makes an incorrect prediction here, and our framework makes the correct one.

4.Synthesis: the reduction of universality (§7): We downgrade "universality" from a fundamental physical law to a statistical approximation valid only in the $L \rightarrow \infty$ limit, and establish "regime boundaries" as the true fundamental quantities of finite systems.

1.4 Structure of the paper

Section 2 details the methodology. Section 3 calibrates on the exact 2D Ising solution and proves the Tool-Dependence Theorem. Section 4 reports the core findings on 3D Ising. Section 5 presents cross-model verification. Section 6 describes the decisive experiment: random-bond Griffiths-phase verification. Section 7 establishes the MRSD tripartite classification and argues for the reduction of universality. Section 8 concludes.

2. Methodology

2.1 The six functional dimensions of the Testability Norms

This paper strictly follows the Testability Norms of Tang (2026g), covering all six functional dimensions (Table 2.1). For the causal-direction dimension, physical reasoning replaces statistical

testing—temperature determines the phase of the system through the Boltzmann factor, its causal status being established by the first principles of statistical mechanics. This is in accord with the norm that "failure of the causal-direction test does not negate the determination".

Table 2.1: Implementation of the six functional dimensions in this paper

Functional dimension	Tools used in this paper	Coverage
Search completeness	Full-range Chow scan + Bai-Perron exhaustive search	All five models
Modulation relation	Interaction-effect test (M–h data)	2D/3D Ising
Structural break	Six-tool cross-validation	All five models
Causal direction	Physical reasoning	Temperature → phase
Problem completeness	5W2H framework	Seven aspects quantified
Process closed-loop	PDCA (calibrate→explore→verify→classify)	Entire workflow

2.2 Six-tool deployment for the structural-break dimension

The structural-break dimension is the core dimension for verifying the systematicity of regime switches. We deploy six independent tools based on different mathematical principles (Table 2.2), covering the full logical chain "known breakpoint → unknown breakpoint → no assumed breakpoint".

Table 2.2: Six tools for structural breaks

Tool	Mathematical principle	Question addressed
Chow test	F-test for equality of regression coefficients across two groups	"Does a break occur at X?"
Threshold regression	Exhaustive candidate breakpoints + Bootstrap confidence interval	"Where is the strongest break, and how wide is the confidence interval?"
SupF test	Bootstrap significance of the maximum Chow F over all candidates	"Does any structural break exist?"
Bai-Perron sequential test	Global RSS minimization + BIC penalty, searching for multiple breakpoints	"How many breakpoints, and where?"
CUSUM test	Cumulative sum of recursive residuals	"Are the regression parameters

Tool	Mathematical principle	Question addressed
		drifting?"
MOSUM test	Moving-window cumulative sum of residuals	"Where does the residual structure become anomalous?"

2.3 Data sources and parameter transparency

Table 2.3: Data files used in this paper

File name	Content	Source
analytic_ising_data.csv	2D Ising M–h response, 1000 points	Onsager-Yang exact formulas
pure_analytic_M_vs_T.csv	2D Ising M–T baseline, 500 points	Onsager-Yang exact formulas
data_coarse_scan.csv	3D Ising L=64 coarse scan, $M_{sp}/\chi/C/U_4$, 120 points	Wolff algorithm (Tang, 2026l)
data_A_L64_full.csv	3D Ising L=64 M–h full data, 81 T × 5 h	Wolff algorithm (Tang, 2026l)
data_fss_L16_32_128.csv	3D Ising FSS multi-size, L=16/32/128	Wolff algorithm (Tang, 2026l)
data_potts_q3_q4.csv	2D Potts q=3/q=4 E_{mean} , L=16/32/64/128	Monte Carlo simulation (Tang, 2026l)
data_XY_KT.csv	2D XY M_{sp} , L=16/32/64/128	Monte Carlo simulation (Tang, 2026l)
random_bond_L16_32.csv	Random-bond 3D Ising, 10% antiferromagnetic bonds, L=16/32	Metropolis algorithm (newly generated for this paper)

The file `random_bond_L16_32.csv` was generated independently for the present work, after the finalization of Tang (2026l). All other data files originate from the simulations of Tang (2026l) and have passed integrity checks (all 10 files have zero missing values, zero duplicate rows, and zero infinite values).

Parameter transparency: The minimum number of points per segment (`min_points`) in the Chow scan is set according to the characteristics of each dataset (Table 2.4). Smaller values allow breakpoints to be detected closer to the edges of the data, potentially capturing earlier signals of structural change, but yield larger estimation variance; larger values provide more stable estimates. In each analysis, the parameter setting is explicitly stated, and we have verified that the core conclusions are insensitive to the choice of parameter (§4.1).

Table 2.4: Chow-scan parameter settings

Dataset	min_points	Rationale
3D Ising coarse scan (§4.1)	8	Balance between search completeness and statistical stability
3D Ising FSS (§4.4)	5	Small samples require a more lenient parameter
Potts $q=4$ (§5.1)	5	Small sample size
Potts $q=3$ (§5.2)	5	Small sample size
XY (§5.3)	4	Very small sample size ($L=128$ has only 16 points)

Honest statement of data limitations: The current data for Potts $q=3$ and the XY model are zero-field simulations, containing only E_{mean} or M_{sp} and lacking the external-field ($M-h$) dimension. Consequently, the modulation-relation dimension of the Testability Norms cannot be verified directly on these models via the interaction-effect test—this is a genuine limitation of the present data, not a methodological flaw. As substitutes, the modulation relation for Potts $q=3$ is indirectly supported by the intercept-jump analysis (§5.2)—the significant difference in intercept across the breakpoint indicates that the regime switch indeed alters the baseline behavior of the executive factor. For the XY model, the modulation relation is indirectly supported by the convergence-trend test (§5.3)—the monotonic convergence of the breakpoint with L is a characteristic finite-size scaling behavior of the KT transition. Direct interaction-effect tests on these models can be performed in future work after external-field response data are generated.

3. Calibration and the Tool-Dependence Theorem

3.1 Calibration on the exact 2D Ising solution

Before deploying the toolchain on systems without analytical solutions, it must first pass calibration on a system with known truth. The 2D Ising model possesses mathematically rigorous exact solutions derived by Onsager (1944) and Yang (1952). The critical temperature is determined by the condition $\sinh(2/T_c)=1$, yielding $T_c=2/\ln(1+\sqrt{2})\approx 2.269185$.

A Chow scan ($\text{min_points}=10$) on the 500 temperature points of `pure_analytic_M_vs_T.csv` automatically locates $T_c=2.273547$, deviating from the exact value by 0.004362 (0.192%). The maximum

Chow F is 8178.6. In 500 random permutations of the temperature labels, not a single false positive of comparable magnitude is produced (empirical $p=0.0000$).

Using the exact $T_c=2.269185$ as the partition, an interaction-effect test is performed on the M - h data of `analytic_ising_data.csv` (modulation-relation dimension). The regression equation is $M \sim h + \text{regime} + h \times \text{regime}$. The interaction coefficient $h \times \text{regime}$ is -0.4739 , with a p -value of zero at the double-precision limit, and $R^2=0.9992$. The temperature regime systematically resets the exposure coefficient of the magnetization to the external field—this is the typical statistical signature of rule-resetting switching, in exact information-theoretic duality with the Ehrenfest second-order transition (Tang, 2026k).

3.2 The mathematical origin of χ -blindness

The Chow behavior of χ is examined on the exact 2D Ising solution (Table 3.1). Near T_c , $\chi(T)$ is a symmetric even function. The Chow test compares the regression coefficients of two groups of data. For $\chi(T)$, the shapes of the curve on either side of T_c are symmetric about T_c ; the slopes on the two sides are equal in magnitude and equal in sign, so the two regressions are nearly identical and the Chow F is close to zero. The residual $F=16.8$ arises from minor asymmetry in the data-point distribution, not from a structural break—confirming that χ is indeed blind to the regime switch. For $M(T)$, the slopes on the two sides have opposite signs (in the ferromagnetic phase M rises as T decreases, while in the paramagnetic phase M falls as T increases), and the Chow test immediately detects this reversal. The function $1/\chi(T)$ forms a V-shaped cusp at T_c —the slope changes abruptly from negative to positive—and the Chow F explodes to $\sim 10^{21}$.

Table 3.1: χ -blindness on the exact 2D Ising solution

Explained variable	Chow F
$M(T)$	10800.0
$\chi(T)$	16.8
$1/\chi(T)$	$\sim 2.2 \times 10^{21}$

This is not a "method misuse"—the calibration on the exact 2D Ising solution has already proved that the Chow test is extremely accurate when paired with the correct mathematical structure. χ tells you "where

the fluctuations are largest"; the Chow test tells you "where the regression equation breaks". These are two distinct physical questions. For fifty years, the method of locating T_c by the χ peak has systematically missed the "regime-switch" dimension revealed by the Chow framework.

3.3 The Tool-Dependence Theorem

On the same exact 2D Ising data, the "critical temperature" is located using the Chow test and the susceptibility-peak method separately:

Table 3.2: Tool dependence on the exact 2D Ising solution

Method	Detected value	Deviation from exact T_c	Direction
Chow test	2.273547	+0.004362	High-T
Susceptibility peak	2.267535	−0.001650	Low-T
Difference between methods	—	0.006012	Straddling the true value

Two widely accepted mathematical methods, applied to a model with an exact analytical solution, yield two different temperatures lying on opposite sides of the true value. This cannot be dismissed as "method error"—if it were mere error, both results would lie on the same side. That they straddle the true value indicates that the two tools are capturing different mathematical dimensions of the phase transition.

On 3D Ising $L=64$, this tool dependence is further amplified: the Chow test gives $T=4.50$, the midpoint of the Bai-Perron double breakpoints gives $T=4.30$, the half-maximum method gives $T=4.40$, and the χ -peak method gives $T=4.50$ —five methods applied across multiple observables yield a T_c range spanning 0.325 temperature units (see §4.1–4.2 for details). These discrepancies are not "pseudo-critical temperatures" on their way to converging to the true value—on the exact 2D Ising solution, none of the methods gives 2.269185.

In a finite system, "critical temperature" is not an objective quantity independent of the detection tool. To claim that "a phase transition occurs at T_c " without specifying "by which tool" is an incomplete proposition. This constitutes the first reduction of the universality concept—the "universal" critical temperature itself is not an objective existence.

4. 3D Ising: Bimodal Splitting, Interaction Effect, and Regime Boundaries

4.1 Discovery of bimodal splitting

A Chow scan ($\text{min_points}=8$) is performed on the four observables of `data_coarse_scan.csv` ($L=64$, 120 temperature points, $T \in [2.0, 8.0]$):

Table 4.1: Four-variable Chow scan for 3D Ising $L=64$ ($\text{min_points}=8$)

Observable	Breakpoint T	Chow F	Physical meaning
M_{sp}	4.50	939.9	Ground-state locking—spontaneous magnetization is fully established
U_4	4.55	299.1	Structural change of fluctuations—Binder cumulant switches
C	4.55	41.1	Peak of energy fluctuations
χ	4.70	13.9	Susceptibility (χ -blindness, see §3.2 and §4.5)

Honest statement of parameter sensitivity: If min_points is reduced to 5 (the setting of Tang, 2026l), the optimal breakpoint for M_{sp} shifts to $T=4.40$ ($F=1524.1$) and for U_4 to $T=4.50$ ($F=2335$), increasing the gap to 0.10. Both settings detect overwhelming signals of regime switching in the interval 4.40–4.55 ($F>900$). The small sensitivity of the breakpoint position to min_points (0.10) is a known statistical property of the Chow test with respect to the minimum segment sample size—a smaller min_points allows breakpoints closer to the data edges and may capture earlier structural-change signals. The blind Bai-Perron double-breakpoint detection (§4.2) is entirely independent of the min_points parameter and independently confirms the bimodal-splitting structure.

The Chow-F peaks of M_{sp} and U_4 do not coincide—they split into two distinct regime boundaries at $T=4.50$ and $T=4.55$ ($\text{min_points}=8$), separated by 0.05. We term this phenomenon **bimodal splitting**: on a finite lattice, the ferromagnetic–paramagnetic transition is not an instantaneous event but splits into two successive regime switches—first, ground-state locking (M_{sp} dimension) at $T=4.50$, then a structural change of fluctuations (U_4 dimension) at $T=4.55$. Between the two switching points lies a **transition regime** of width ≈ 0.05 —the old equation has collapsed but the new one is not yet fully established. In standard finite-size scaling (FSS) practice, this region is usually regarded as "a rounding part to be extrapolated away". Here we find that it is an independent statistical structure with sharp boundaries that can

be automatically identified by an algorithm. In the thermodynamic limit ($L \rightarrow \infty$), the transition regime collapses to zero and the two switches merge into the classical single transition. On the finite lattice $L=64$, however, it unfolds into a real, independent regime.

4.2 Structural breaks: six-tool cross-validation and robustness

All six structural-break tools are deployed on the 3D Ising M_{sp} data (structural-break dimension):

Table 4.2: Six-tool cross-validation for 3D Ising M_{sp}

Tool	Optimal breakpoint	Statistic	Significant?
Chow test	$T=4.50$	$F=939.9$	✓
Threshold regression (Bootstrap 500)	Median=4.50	95% CI=[4.50, 4.65]	✓
SupF test	$T=4.50$	SupF=939.9, Bootstrap $p=0.0000$	✓
Bai-Perron double (intercept series)	$T_1=4.41, T_2=4.56$	F-ratio=4.16, BIC improvement +124.14	✓
Bai-Perron double (slope series)	$T_1=4.42, T_2=4.57$	F-ratio=1.20, BIC improvement +55.20	✓
CUSUM	—	max=33.60, exceeds 5% critical line	✓
MOSUM	—	max=8.09, critical value 2.0	✓

Six independent tools based on different mathematical principles all detect significant structural breaks on 3D Ising. The Bai-Perron double-break gap (~ 0.15) is larger than the observed M_{sp}/U_4 splitting gap (0.05) because Bai-Perron identifies the onset and offset boundaries of the change in the regression equation from the M - h coefficient series, rather than the extremum position of a single observable. The two methods are mathematically independent yet lead to the same conclusion.

Before announcing a discovery, alternative explanations must be excluded. Table 4.3 reports robustness checks for the M_{sp} breakpoint.

Table 4.3: Robustness of the 3D Ising M_{sp} breakpoint

Test	Result	Alternative explanation excluded
Permutation test (500 times)	Empirical $p=0.0020$	Statistical artifact
Noise injection ($\sigma=0.05$)	T unchanged, F=813.3	Over-sensitivity to noise
Random 80% subsample	T unchanged, F=687.1	Driven by a single temperature point
Random 60% subsample	T=4.60, F=804.7	Robust for most of the data
Trim extreme T ($T \in [3,6]$)	T unchanged, F=296.6	Driven by far-ferromagnetic or far-paramagnetic regions

The breakpoint position is extremely robust against noise, data subsampling, and trimming of extreme temperatures.

4.3 Interaction effect: confirmation of rule-resetting type

Using the M_{sp} breakpoint $T=4.50$ as the partition, an interaction-effect test (modulation-relation dimension) is performed on the full M – h data of `data_A_L64_full.csv`:

Table 4.4: Interaction-effect test on 3D Ising (partition $T=4.50$)

Coefficient	Estimate	p-value	Physical meaning
const	0.5575	~ 0	Spontaneous magnetization in the ferromagnetic phase
h	−1.0996	1.20×10^{-35}	Exposure coefficient to h in the paramagnetic phase
regime	−0.5368	2.28×10^{-181}	Intercept drop across the regime
$h \times \text{regime}$	2.1668	0.00e+00	Jump in the exposure coefficient

The interaction term has a p-value of zero at the double-precision limit, with $R^2=0.8874$. The exposure coefficients to the external field in the ferromagnetic and paramagnetic phases are +1.0745 and −1.0996, respectively—opposite in sign and nearly equal in magnitude. This is the characteristic statistical signature of rule-resetting switching, fully consistent with the information-theoretic duality on the exact 2D Ising solution (Tang, 2026k) and corresponding to an Ehrenfest second-order transition.

4.4 The precursor regime boundary

Chow scans ($\text{min_points}=5$) are performed on the M_{sp} data for $L=16, 32$, and 128 from `data_fss_L16_32_128.csv`, and combined with the $L=64$ data ($\text{min_points}=5$) from Tang (2026l).

Table 4.5: Cross-size breakpoints of 3D Ising M_{sp}

L	M_{sp} breakpoint T	Chow F
16	4.55	68.6
32	4.55	115.8
64	4.40*	1524.1
128	4.50	26.1

* $L=64$ data from Tang (2026l) with $\text{min_points}=5$.

The four breakpoints have a mean of 4.50 and a standard deviation of only 0.071. Regression on $\log(L)$ gives a slope of -0.0258 with $p=0.2123$. Statistically, this p -value means that the null hypothesis of "no convergence trend" cannot be rejected—but failure to reject the null hypothesis does not prove it. With only four data points ($L=16, 32, 64, 128$), the degrees of freedom are limited, and it is not yet possible to statistically distinguish "slow convergence" from "structural non-convergence".

It should be honestly noted that the Chow F for $L=128$ (26.1) is significantly lower than those for $L=32$ (115.8) and $L=64$ (1524.1), so the uncertainty of the $L=128$ breakpoint is relatively large. For reference, the Bootstrap threshold regression on the $L=64$ coarse-scan data ($\text{min_points}=8$) gave a 95% CI of $[4.50, 4.65]$, a width of 0.15. $L=128$ has only 16 temperature points, and its standard error is expected to be even larger. The statistical argument for the precursor regime boundary does not hinge on the exact position of any single point—even if the $L=128$ breakpoint were 4.50 ± 0.10 , the mean of the four points would remain ≈ 4.52 and the standard deviation would remain on the order of 0.1.

We term this phenomenon the **precursor regime boundary**—"precursor" not in the temporal sense of a warning signal before the transition, but in the structural sense of a regime boundary that precedes the thermodynamic singularity and remains independent of it. **As a contrast, the XY model with the same four points yields $p=0.0000$ (§5.3).** The difference in size-sensitivity between the two transition types is sharp and robust. Over a 512-fold increase in volume, the M_{sp} breakpoint of a second-order transition

barely moves, while that of a KT transition converges strictly monotonically toward the known T_c . Data at $L \geq 256$ are necessary to distinguish "slow convergence" from "structural non-convergence".

4.5 χ -blindness reproduced on 3D Ising

On the 3D Ising $L=64$ coarse-scan data, χ yields a Chow F of only 13.9 (see Table 4.1)—almost negligible compared to 939.9 for M_{sp} and 299.1 for U_4 . Transforming to $1/\chi$ raises the Chow F to 559.9, an activation factor of 40. The same phenomenon is reproduced on all three sizes of the FSS data: $L=16$ gives a factor of 4, $L=32$ a factor of 19, and $L=128$ a factor of 6. χ -blindness is an intrinsic feature of finite systems that does not disappear with increasing L . Its mathematical origin is given in §3.2.

5. Cross-Model Universality Verification

5.1 Potts $q=4$: cross-size reproduction of bimodal splitting

Bai-Perron double-breakpoint searches are performed on the E_{mean} data for $q=4$ in `data_potts_q3_q4.csv` at $L=16, 32, 64$, and 128 :

Table 5.1: Cross-size double-breakpoint analysis for Potts $q=4$

L	Single-break T	Double-break T_1	Double-break T_2	Gap	F-ratio	BIC improvement	Significant?
16	1.000	0.940	1.080	0.140	1.28	+17.86	✓
32	0.980	0.900	1.080	0.180	3.16	+35.98	✓
64	1.080	0.840	1.080	0.240	—	+6.51	✓ (BIC)
128	1.080	0.940	1.060	0.120	—	−1.35	✗

Double breakpoints receive statistical support on all three sizes $L=16, 32$, and 64 ($F\text{-ratio}>1$ or $BIC\text{ improvement}>0$). $L=128$ is the only size where the double breakpoints are not significant ($BIC\text{ improvement}=-1.35$), fully consistent with the finding of Tang (2026l)—the double-peak gap shrinks from 0.240 at $L=64$ to 0.120 at $L=128$, and the BIC penalty of the double-breakpoint model outweighs its advantage in residual reduction. This is a manifestation of finite-size scaling: as $L\rightarrow\infty$, the transition regime collapses to zero and bimodal splitting merges into the classical single transition.

The E_{mean} breakpoints of Potts $q=4$ across $L=16\text{--}128$ give a $\log(L)$ regression $p=0.12$, consistent with 3D Ising—insensitive to L , not supporting a convergence trend. The precursor regime boundary is independently reproduced on a second second-order model.

5.2 Potts $q=3$: discovery of mixed-type switching

Chow scans and intercept-jump analysis are performed on the E_{mean} data for $q=3$:

Table 5.2: Intercept-jump analysis for Potts $q=3$

L	Single-break T	Chow F	Intercept jump	Slope change	Ratio
16	1.100	87.5	0.0629	0.0574	1.1
32	1.080	240.0	0.0609	0.0569	1.1
64	1.200	53.3	0.1028	0.0699	1.5
128	1.180	1.2	0.2547	0.1963	1.3

The intercept/slope ratio lies between 1.1 and 1.5 —indicating both an intercept jump (characteristic of a first-order transition) and a slope change (characteristic of a second-order transition). On a finite system, a first-order transition is not a pure "direction-reversal type" (pure intercept jump) but rather **mixed-type switching**. For $L=128$, the Chow F is only 1.2 and the break itself is not significant; its ratio is shown only for reference. The ratio does not exhibit a clear monotonic trend across the four sizes, suggesting that, over the accessible size range, the mixed type may be an intrinsic finite-size feature of Potts $q=3$, not necessarily converging to a pure direction-reversal type.

The criterion for mixed-type switching is an intercept/slope ratio <2 together with a significant Chow test. The choice of 2 as the threshold is motivated as follows: in the thermodynamic limit of a first-order transition, the latent heat is fully developed, the intercept jump tends to infinity, and the ratio tends to ∞ ; in

the ideal case of a second-order transition, the intercept is continuous and the ratio tends to 0. A ratio <2 means that the intercept jump and the slope change are of comparable magnitude—both make substantial contributions to the regime switch, being neither purely intercept-dominated (ratio $\gg 2$, pure direction-reversal) nor purely slope-dominated (ratio $\ll 1$, pure rule-resetting). The value 2 is a pragmatic criterion; its exact value can be calibrated as data on more models accumulate.

The physical significance of this finding is that the idealized first-order (pure intercept jump) and second-order (pure slope switch) transitions are not sharply distinct in finite systems—they lie at opposite ends of a continuous spectrum, and the intercept/slope ratio is a quantitative index characterizing this spectrum.

5.3 XY model: single-stage continuous crossover

Chow scans are performed on the M_{sp} data of data_XY_KT.csv:

Table 5.3: Cross-size analysis for the XY model

L	Single-break T	Chow F	Deviation (from $T_c=0.8929$)	BP double	CUSUM	Precursor boundary
16	0.750	84.1	0.143	BIC<0	✗	—
32	0.800	12.6	0.093	BIC<0	✗	—
64	0.850	13.1	0.043	BIC<0	✗	—
128	0.900	1.8	0.007	BIC<0	✗	—

Three features contrast sharply with second-order transitions: (1) the deviation converges strictly monotonically ($0.143 \rightarrow 0.093 \rightarrow 0.043 \rightarrow 0.007$), perfectly embodying the FSS convergence law; (2) Bai-Perron double breakpoints are consistently not significant (BIC<0); (3) CUSUM is consistently not significant. Regression on $\log(L)$ gives $p=0.0000$ —an extremely significant convergence trend.

Fitting the four data points to $T_c(L)=T_c(\infty)+A \cdot L^{-1/\nu}$ (with $\nu=0.67$ for the XY model) gives an extrapolated $T_c(\infty) \approx 0.894 \pm 0.006$, consistent with the known KT temperature 0.8929 ($R^2=0.98$).

The XY model exhibits no bimodal splitting, no precursor regime boundary, and no significant CUSUM—it accomplishes the regime switch through a single-stage continuous crossover, forming a clear tripartition with first-order and second-order transitions.

6. Decisive Experiment: Random-Bond Griffiths-Phase Verification

Traditional universality theory predicts that, as long as the disorder concentration is below the percolation threshold, a pure model and a disordered model belong to the same universality class—only the critical exponents are modified. The Factor Hierarchy Law, however, predicts that disorder will generate Griffiths phases—rare, unfrustrated ferromagnetic clusters that freeze locally before the global phase transition. The freezing of these clusters produces additional variance structures that Bai-Perron will detect as independent regime boundaries absent from the pure model. **This is the key point of divergence between the two theories.**

6.1 Simulation design

On a 3D cubic lattice, 10% of the nearest-neighbor bonds are randomly set to antiferromagnetic ($J=-1$); the rest remain ferromagnetic ($J=+1$). Simulations are performed for $L=16$ and 32 , $T \in [4.0, 5.2]$, step 0.05 , 25 temperature points. The Metropolis algorithm (which supports mixed bonds) is used, with 1000 thermalization steps and 3000 production steps. Four-thread parallelism is employed, saving results as they are generated, with checkpoint-resume capability. The data are saved as `random_bond_L16_32.csv`. This file was generated independently for this paper after the finalization of Tang (2026l).

6.2 Analysis results

Table 6.1: Six-tool analysis of the random-bond 3D Ising model

L	Variable	Chow peak T	Chow F	B P T ₁	B P T ₂	F-ratio	BIC improvement	Double?	CUSUM	MOSUM	Per m. p
16	M	4.50	26.4	4.25	4.50	0.60	+5.25	✓	✓	✓	0.0000
16	E	4.50	73.8	4.50	4.75	0.24	-1.08	✗	✓	✓	0.0000
32	M	4.50	37.1	4.25	4.50	0.56	+4.66	✓	✓	✓	0.0000

L	Variable	Chow peak T	Chow F	B P T ₁	B P T ₂	F-ratio	BIC improvement	Double?	CUSUM	MOSUM	Per m. p
32	E	4.55	79.8	4.25	4.75	6.16	+42.77	✓	✓	✓	0.0000

Core finding: the systematic appearance of a new breakpoint at $T \approx 4.25$. On the pure 3D Ising model, no structural break has ever been detected near $T \approx 4.25$ —the M_{sp} breakpoint lies at 4.40–4.50 and U_4 at 4.50–4.55; $T \approx 4.25$ is completely "clean". On the random-bond model, $T \approx 4.25$ is detected as the first breakpoint by Bai-Perron in three of the four independent analyses ($L=16$ M, $L=32$ M, $L=32$ E). For $L=32$ E, the BIC improvement reaches +42.77 with an F-ratio of 6.16—overwhelming statistical evidence.

In some analyses the F-ratio is <1 while the BIC improvement is >0 (e.g., $L=16$ M: F-ratio 0.60, BIC improvement +5.25; $L=32$ M: F-ratio 0.56, BIC improvement +4.66). In small samples, BIC is a more reliable model-selection criterion than the F-ratio because it simultaneously accounts for goodness-of-fit and parameter penalty. Three of the four analyses have positive BIC improvements, consistently supporting the double-breakpoint model.

Physical interpretation: $T \approx 4.25$ is identified as the statistical signature of the Griffiths phase. In a disordered system, a few ferromagnetic clusters that are not destroyed by antiferromagnetic bonds have a local coupling strength higher than the global average. These clusters undergo local freezing before the global phase transition ($T \approx 4.50$). This "premature freezing" produces an additional variance structure that is detected by the Chow framework as an independent regime boundary. The region between the two breakpoints (4.25–4.50) constitutes the Griffiths regime—an intermediate phase in which some clusters have already frozen but global order has not yet been established. The uniform significance of CUSUM and MOSUM indicates that disorder systematically alters the statistical properties of the residuals over the entire temperature axis.

6.3 Comparison with the pure model

Table 6.2: Regime-boundary comparison: pure vs. random-bond 3D Ising

Feature	Pure 3D Ising	Random-bond 3D Ising
Number of breakpoints	2 ($M_{\text{sp}} + U_4$)	3 (Griffiths + $M_{\text{sp}} + E/U_4$)
First breakpoint	None	$T \approx 4.25$ (Griffiths phase)
Main breakpoint	$T \approx 4.50$	$T \approx 4.50$ (preserved)
CUSUM/MOSUM	Partially significant	Uniformly significant
Transition regime	4.40–4.49 (pure model)	4.25–4.50 (Griffiths-extended)

Traditional universality classes lump the pure and disordered models together. The topology of regime boundaries reveals fundamentally different structures in the two systems. The label "universality class" erases real, algorithmically detectable structural differences that exist in finite systems.

If $T \approx 4.25$ is indeed the statistical signature of the Griffiths phase, then its sensitivity to the disorder concentration is testable: lowering the disorder concentration should shift $T \approx 4.25$ to higher temperatures or eliminate it altogether; raising the disorder concentration should expand the Griffiths region. This is a testable prediction left for future work.

This result constitutes decisive evidence for a paradigm shift—where traditional theory makes an incorrect prediction (that pure and disordered models are equivalent in the long-wavelength limit), the Factor Hierarchy Law makes the correct prediction (that disorder generates additional regime boundaries).

7. Discussion

7.1 The MRSD tripartite classification framework

Based on the cross-model evidence, we establish an MRSD tripartite classification of finite-system phase transitions (Table 7.1). In Tang (2026l), MRSD was positioned as a universal algorithm engine. Building on that positioning, the present work further applies it to a tripartite classification framework for finite-system phase transitions. This framework does not rely on any prior definition of a free-energy function, Hamiltonian, or order parameter—it depends only on whether the data are objective.

Table 7.1: MRSD tripartite classification of finite-system phase transitions

Transition type	Representative models	Bimodal splitting	Precursor boundary	BP double	CUSUM	FSS convergence	Switching form
First-order	Potts $q=3$	No	No	Mixed	Significant	TBD	Mixed-type
Second-order	3D Ising, Potts $q=4$	Yes	Yes	Significant	Significant	No	Rule-resetting + bimodal
KT	XY	No	No	Not significant	Not significant	Yes	Single-stage continuous

Formal criteria:

- Bai-Perron double breakpoints significant ($F\text{-ratio} > 1$ or $BIC > 0$) + CUSUM significant + precursor regime boundary present → **Rule-resetting + bimodal splitting (second-order)**
- Bai-Perron double breakpoints not significant ($BIC < 0$) + CUSUM not significant + deviation converges monotonically → **Single-stage continuous crossover (KT)**
- Intercept jump and slope change of comparable magnitude (intercept/slope ratio < 2) with significant Chow test → **Mixed-type switching (first-order on finite system)**

The criterion for the mixed type (ratio < 2) is based on the following reasoning: in the thermodynamic limit of a first-order transition, the latent heat is fully developed, the intercept jump tends to infinity, and the ratio tends to ∞ ; in the ideal second-order case, the intercept is continuous and the ratio tends to 0. A ratio < 2 indicates that the two contributions are of comparable magnitude and both contribute substantially to the regime switch. The value 2 is a pragmatic criterion whose exact value can be refined as data on more models accumulate.

7.2 The reduction of universality

The following five propositions argue, from different angles, a single core conclusion: **Traditional universality is a mathematical convergence product of the thermodynamic limit ($L \rightarrow \infty$). In real finite**

systems, regime boundaries are more fundamental physical quantities than universality classes. This paradigm shift, from "universality" to "regime boundaries", constitutes a "reduction of universality" within the traditional theory of phase transitions—universality is downgraded from a fundamental physical law to a statistical approximation valid only in the $L \rightarrow \infty$ limit.

Proposition 1 (A phase transition is a zone, not a point). In a finite system, different physical quantities complete their regime switches at different temperatures. On 3D Ising $L=64$, M_{sp} locks at 4.50, U_4 changes structure at 4.55, and χ peaks at 4.70. They would converge to a single point only in the thermodynamic limit. In any real system, they separate into a regime-boundary spectrum. A phase transition is not an instantaneous event—it is a process with internal structure.

Proposition 2 (The detection tool defines the "critical point"). There is no "correct" detection method. On the exact 2D Ising solution, two methods give temperatures straddling the true value. On 3D Ising $L=64$, five methods across multiple observables yield a T_c range of 0.325 temperature units. To claim that "a phase transition occurs at T_c " without specifying "by which tool" is an incomplete proposition.

Proposition 3 (Finite systems possess structural boundaries independent of the thermodynamic limit). The precursor regime boundary is extremely insensitive to L in 3D Ising (SD 0.071) and Potts $q=4$ ($p=0.12$), in sharp contrast to the significant convergence of the KT transition ($p=0.0000$). A finite system is not a defective copy of the thermodynamic limit—it possesses real, scale-invariant structures of its own.

Proposition 4 (χ -blindness is a mathematical necessity, not a method misuse). The mathematical mismatch between the even-function structure of χ and the odd-function requirement of the Chow test causes χ to be systematically silent in the face of structural-break detection. The activation of $1/\chi$ is the inevitable inversion of this mathematical structure. χ tells you "where the fluctuations are largest"; the Chow test tells you "where the regression equation breaks"—these are two different physical questions.

Proposition 5 (Disorder generates extra regime boundaries—and traditional theory fails here). The random-bond model produces a new Griffiths-phase breakpoint ($T \approx 4.25$) beyond the regime boundaries of the pure model. Traditional universality classes lump the two together, but the regime-boundary topology reveals fundamentally different structures. This is decisive evidence for a

paradigm shift—where traditional theory makes an incorrect prediction, the Factor Hierarchy Law makes the correct one.

7.3 A modified finite-size scaling ansatz

The standard FSS ansatz assumes that all pseudo-critical temperatures converge to the thermodynamic limit according to $T_c(L) - T_c(\infty) \propto L^{-1/\nu}$. The discovery of the precursor regime boundary demands a modification. For regime boundaries detected by structural-break tools, the size dependence obeys:

$$B(L) = T_c(\infty) \pm \Delta T_0 \cdot \Phi(L)$$

where $\Phi(L)$ is a scaling function. For the M_{sp} dimension, over the intermediate range $L=16-128$, $\Phi(L) \approx \text{constant}$ —the system possesses an intrinsic, scale-invariant structural boundary over this range. For pseudo-critical temperatures detected by traditional extremum methods (χ peak, Binder-cumulant crossing), $\Phi(L)$ reverts to the standard power-law form.

This modification redefines "finite-size effects" as independent statistical structures with sharp boundaries, rather than "errors to be extrapolated away". This is a fundamental revision of FSS by the Factor Hierarchy Law—finite-size broadening is not an error, but a physical reality.

7.4 Complete closure of the Testability Norms

Table 7.2: Verification status of the six functional dimensions

Functional dimension	Tool coverage	Core result	Status
Search completeness	Full-range Chow scan + Bai-Perron exhaustive search	All five models covered	✓
Modulation relation	Interaction-effect test	2D/3D Ising: $p=0.00e+00$	✓
Structural break	Six-tool cross-validation	Second-order: all passed; first-order and KT: partially passed	✓
Causal direction	Physical reasoning	Temperature $\rightarrow$ phase	✓

Functional dimension	Tool coverage	Core result	Status
Problem completeness	5W2H	All seven aspects quantified	✓
Process closed-loop	PDCA	Calibrate→explore→verify→classify	✓

7.5 Five testable challenges

We issue the following testable challenges to the traditional theory of phase transitions:

1.Tool-dependence challenge: On any finite-size 3D Ising model, locate T_c using the Chow test, the χ -peak method, and the Binder-cumulant crossing method separately. We predict that the three methods will yield three distinct temperatures, and that the differences will not vanish with increasing L .

2.Bimodal-splitting challenge: On the 2D Potts $q=4$ model, we predict that the breakpoints of the M_{sp} and U_4 dimensions will be separated and that the M_{sp} breakpoint will not converge with L .

3.Symmetry challenge: Test whether the precursor regime boundary persists on the 3D Heisenberg model ($O(3)$ symmetry). If it does, the precursor regime boundary is a universal feature of continuous transitions, independent of symmetry. If it does not, it is confined to discrete-symmetry systems.

4.Non-equilibrium challenge: Deploy the MRSD algorithm on any non-equilibrium phase-transition system (e.g., the DP model). We predict that Bai-Perron will automatically identify the regime-switch morphology.

5.Experimental-data challenge: Run the Chow test on the magnetization curves of real magnetic materials. We predict that a bimodal-splitting structure consistent with the simulation data will be detected.

8. Conclusion

This paper has performed a systematic cross-model analysis on five phase-transition models using the Factor Hierarchy Law and the Testability Norms. **Five core conclusions:**

(1) Bimodal splitting is a universal structure of second-order transitions on finite lattices. This means that, in a finite system, a phase transition is not an instantaneous event—it is a process with internal structure, consisting of two successive stages (ground-state locking and fluctuation-structure change) separated by an independent transition regime. On 3D Ising $L=64$, M_{sp} breaks in the interval $T \approx 4.40\text{--}4.50$ ($F=939.9\text{--}1524.1$) and U_4 breaks in the interval $T \approx 4.50\text{--}4.55$ ($F=299.1\text{--}2335$). The same structure is reproduced on Potts $q=4$ for $L=16, 32$, and 64 . Six-tool cross-validation is unanimously passed.

(2) The precursor regime boundary demonstrates that finite systems possess structural boundaries independent of the thermodynamic limit. The M_{sp} breakpoint is extremely insensitive to size in 3D Ising ($L=16\text{--}128$, SD 0.071) and Potts $q=4$ ($L=16\text{--}128$, $p=0.12$). The KT transition (XY model) lacks this boundary ($p=0.0000$). The available data are not yet sufficient to strictly distinguish "slow convergence" from "structural non-convergence", but the sharp contrast in size-sensitivity between the two transition types already constitutes a robust empirical regularity.

(3) The Tool-Dependence Theorem has received decisive proof. On the exact 2D Ising solution, the Chow test and the susceptibility-peak method give "critical temperatures" straddling the true value. No "critical temperature" exists independently of the detection tool—fifty years of discussion about T_c has essentially been a discussion about the operation of detection tools.

(4) χ -blindness reveals a systematic blind spot of the traditional gold standard. The root cause is a mathematical mismatch between the even-function structure of χ and the odd-function requirement of the Chow test. $1/\chi$ provides the mathematically correct alternative.

(5) The random-bond Griffiths-phase verification is decisive evidence for a paradigm shift. Traditional universality lumps pure and disordered models together, but the Factor Hierarchy Law detects, on the disordered model, a new breakpoint at $T \approx 4.25$ that is absent from the pure model—where traditional theory makes an incorrect prediction, we make the correct one.

Overall conclusion: Traditional universality is a mathematical convergence product of the thermodynamic limit. In real finite systems, regime boundaries are more fundamental physical quantities than universality classes. This paradigm shift, from "universality" to "regime boundaries", constitutes a "reduction of universality" within the traditional theory of phase transitions. The Factor Hierarchy Law and the Testability Norms constitute a complete theoretical framework for phase

transitions in finite systems—relying on neither a free-energy function, nor a Hamiltonian, nor a prior definition of an order parameter, but only on whether the data are objective.

Data Availability Statement

All data files used in this paper are archived at D:\学术\第十三篇物理学论文数据, with the file list given in Table 2.3 of §2.3. All data files have passed integrity checks (zero missing values, zero duplicate rows, zero infinite values across all 10 files). The 2D Ising data were generated from the Onsager-Yang exact formulas. The 3D Ising data were generated using the Wolff cluster algorithm (Tang, 2026l). The random-bond 3D Ising data were generated using the Metropolis algorithm specifically for this paper. All analysis code is available upon request.

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